

Lost in Translation: An Actor-Network Theory Analysis of NEPA's Permitting Network

STS Research Paper

4920 Words

A Thesis

In STS 4600

Presented to

The Faculty of the

School of Engineering and Applied Science

University of Virginia

In Partial Fulfillment of the Requirements for the Degree
Bachelors of Science in Systems and Information Engineering

By

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May 5th 2026

On my honor as a university student, I have neither given nor received unauthorized aid on this assignment as defined by the honor guidelines for thesis related assignments.

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I. Introduction

The United States is facing two simultaneous and ostensibly competing imperatives: accelerating the deployment of clean energy infrastructure, and ensuring that the communities most burdened by that infrastructure retain meaningful influence over decisions that affect them. At the center of both is the National Environmental Policy Act (NEPA) of 1969—a statute that has been called the "Magna Carta" of federal environmental law and, with roughly equal conviction, a bureaucratic obstacle to the energy transition. This paper argues that both characterizations miss what is most important about NEPA: not whether it is fast or slow, but what kind of knowledge it legitimizes, whose interests it enrolls, and which communities it structurally excludes.

To examine these questions, this paper applies two analytical frameworks from Science, Technology, and Society (STS) scholarship to NEPA's permitting network. The first is Actor-Network Theory (ANT), developed by Bruno Latour, Michel Callon, and John Law, which treats both human and non-human entities as actants whose interactions assemble, stabilize, or destabilize sociotechnical networks. Applied to NEPA, ANT reveals that the permitting process is not a neutral administrative procedure but a network-building exercise in which federal agencies, EIS documents, pipeline routes, courts, developers, Indigenous nations, and financial instruments all play constitutive roles. The second framework is environmental justice, specifically its three-dimensional formulation encompassing procedural justice (fairness of process), distributive justice (equity of outcomes), and recognition justice (genuine acknowledgment of marginalized communities' values and worldviews). Together, these frameworks make visible what a purely legal or policy analysis would obscure: that NEPA's failures are not primarily technical but structural, embedded in the design of the network itself.

The paper proceeds in four parts. It first situates NEPA historically and legally, tracing its origins in the environmental crises of the late 1960s, its evolution from a substantive environmental protection tool to a purely procedural statute, and the key regulatory milestones that have reshaped its requirements. It then introduces ANT and environmental justice as analytical lenses. The paper applies these frameworks to two case studies: the Dakota Access Pipeline, in which the USACE's scoping process functionally excised the Standing Rock Sioux Tribe from the permitting network despite their formal participation; and the Mountain Valley Pipeline provision of the FRA, in which Congress acted as a macro-actant to override years of appellate reversals by forcibly re-enrolling the entire permitting network through debt-ceiling legislation. Finally, the paper examines the permitting reform debate, arguing that the dominant narrative—that NEPA is the primary obstacle to clean energy deployment—is empirically unsupported and displaces conflict over infrastructure costs onto procedural processes rather than addressing the actual binding constraints.

The central argument of this paper is that NEPA's deepest problem is not inefficiency but a structural translation failure: the permitting network it produces cannot accommodate the

worldviews, values, and treaty-based relationships of the communities most likely to bear the burdens of energy infrastructure, and recent reform efforts have made this problem worse rather than better.

II. Background: NEPA as a Sociotechnical Artifact

A. Origins and Purpose (1969–1970)

NEPA was first signed into law on January 1, 1970 by President Richard Nixon, and is often referred to as the "Magna Carta" of federal environmental policy. Public urgency for environmental protection arose largely in the wake of two high-profile disasters: the Santa Barbara oil spill and the Cuyahoga River fire, both occurring in 1969. These crises alerted the public to the government's shortcomings in addressing air and water pollution, haphazard land use, and the deterioration of natural systems, reflecting a national consensus that prior federal law had largely ignored citizens' ecological values (NEPA; Cram et al.).

Legally, NEPA serves two distinct goals: informing the agency decision-maker and informing the public, with public disclosure carrying independent value regardless of whether it shapes the agency's final choice (*Robertson v. Methow Valley Citizens Council*). NEPA requires that "major federal actions significantly affecting the quality of the human environment" undergo formal environmental review. When a project's impacts are known to be significant, compliance requires completion of an Environmental Impact Statement (EIS): the lead agency publishes a Notice of Intent, prepares a Draft EIS analyzing direct, indirect, and cumulative impacts and alternatives including a "no action" baseline, holds a public comment period, and issues a Final EIS and Record of Decision. Because NEPA requires public disclosure and opens governmental decisionmaking to scrutiny, it also exerts a preventative influence on agency action: agencies face internal and external pressures to design projects with reduced adverse environmental consequences to avoid damaging disclosures. Bradley Karkkainen characterizes NEPA as among the most democratic environmental statutes, reasoning that its public participation process "forces the public and the government officials to interface," though he cautions that organized NGOs are better positioned than ordinary citizens to exploit the comment process, meaning NEPA tends to advance interest group pluralism more than direct democracy (Karkkainen 913).

B. How NEPA Works: Categorical Exclusions, Environmental Assessments, and EIS

NEPA's implementing regulations establish a 3-tiered review framework calibrated to the expected severity of a proposed action's environmental impacts. Categorical Exclusions (CEs)—covering approximately 94% of actions subject to NEPA—apply to actions that do not individually or cumulatively produce significant environmental effects and require no further analysis unless "extraordinary circumstances" exist. Environmental Assessments (EAs), covering roughly 5% of actions, determine whether impacts are insignificant (resulting in a Finding of No Significant Impact, or FONSI) or significant enough to warrant a full EIS. The EIS tier, reserved

for actions with expected significant environmental effects, constitutes a major technical undertaking requiring analysis of direct, indirect, and cumulative impacts and multiple rounds of public comment (Mortimer et al.; Scott et al.). EISs account for only about 1% of all NEPA actions, and annual issuances have fallen substantially from over 2,000 in the 1970s to fewer than 200 in the 2020s—a fact notably absent from most reform debates (Ruhl & Buzbee).

C. Evolving Interpretations and Legislative Changes

When NEPA was enacted, courts read it as a substantive law, imposing genuine obligations to protect the environment. In *Calvert Cliffs' Coordinating Committee v. AEC* (1971), a federal appeals court held that agencies had to carry out NEPA's duties "to the fullest extent possible." By 1989, however, the Supreme Court settled in *Robertson v. Methow Valley Citizens Council* that "NEPA itself does not mandate particular results, but simply prescribes the necessary process," converting the statute from a substantive tool to a procedural one. An agency can now acknowledge serious environmental harm in an EIS and still proceed, as long as the required analysis was conducted (Karkkainen).

In 1978, the Council on Environmental Quality (CEQ) issued binding implementing regulations aimed at saving time and reducing paperwork, responding to President Carter's concern that EISs had become unwieldy documents measured by volume rather than usefulness. The 2020 Trump-era revisions imposed page limits, narrowed the definition of actionable impacts, and restricted alternatives to those "technically and economically feasible." In 2023, the Fiscal Responsibility Act (FRA) enacted the first statutory amendments to NEPA since its original passage, codifying elements of the 2020 regulations, adding review timelines, and introducing a "One Federal Decision" streamlining framework (Clark, Kanatas & Smith). Scholarly opinion on the FRA is divided: some view the changes as modest codifications of long-standing practice, while critics argue they quietly eliminate NEPA's public disclosure purpose by narrowing agency review obligations to focus almost exclusively on the decision-maker rather than broader transparency—reflecting, in their view, a partisan framing of environmental review as bureaucratic "red tape" rather than a meaningful democratic check (Clark, Kanatas & Smith).

III. Analytical Frameworks

A. Actor-Network Theory

Actor-Network Theory is a foundational STS framework developed in the 1980s by Bruno Latour and colleagues including Michel Callon and John Law. ANT asks how humans and non-humans fit together into larger, relatively stable networks that construct social and material reality. Any entity that acts upon or is acted upon within a network is an "actant," and both human and non-human actants are treated as analytically equal contributors to how that reality is built.

The process by which a network is assembled is called translation, which proceeds through four overlapping stages. A focal actor first defines a problem in a way that makes its own involvement seem necessary—problematization. Through interessement, that actor draws other actants in by offering a shared goal or position in the emerging network. When those actants accept their roles, the network has achieved enrollment. Finally, mobilization occurs when the network stabilizes enough that the focal actor can speak and act on behalf of the whole. The point through which all actants must pass to achieve their goals is the Obligatory Passage Point (OPP); controlling the OPP is how a focal actor consolidates network influence (Callon, 1986, as discussed in Aka; Chu et al.).

ANT is well suited to analyzing the NEPA permitting process because it treats the social and the material as equally constitutive of governance outcomes. Chu et al.'s application of ANT to China's carbon governance framework demonstrates that even a molecule—CO₂—can function as a non-human actant shaping human actors' interests when present on a policy agenda, a logic that applies directly to how pipeline routes and EIS documents become sites of network-building and contestation in U.S. permitting disputes (Chu et al. 1). The NEPA permitting network assembles federal agencies, developers, tribal nations, courts, and infrastructure in precisely the heterogeneous arrangement ANT is designed to trace.

B. Environmental Justice Frameworks

Environmental justice encompasses three analytically distinct dimensions. Procedural justice concerns whether a decision-making process is transparent, accountable, and gives genuine agency and voice to affected communities. Distributive justice concerns whether the outcomes of decisions are equitable—whether environmental harms and benefits are fairly distributed and whether historically marginalized groups bear a disproportionate share of negative impacts. As Ulibarri, Pérez Figueroa, and Grant note, distributive justice "is the heart of the early EJ movement, which called attention to the fact that polluting industrial activities and hazardous waste facilities tended to be disproportionately located in low-income, minority, and other politically-disempowered communities" (Ulibarri et al. 2). A process can satisfy every procedural requirement while still producing deeply inequitable outcomes—these two dimensions are distinct and non-interchangeable goals.

Recognition justice, the third dimension, argues that a just process must go beyond formal participation to genuinely acknowledge and respect the worldviews and values that affected communities bring, particularly in contexts of historic marginalization where procedural access alone may not produce meaningful voice. It is often described as the bridge between procedural and distributive justice: without first recognizing the power dynamics that shape who participates and on whose terms, formally open processes risk replicating inequitable distributional outcomes. Recognition justice raises the question of "who defines what is just, and for whom" (Newell and Mulvaney, cited in Ulibarri et al. 8), making it especially relevant to NEPA contexts

where technical document formats and short comment windows can systematically exclude the communities most likely to bear a project's burdens.

C. Democratic Legitimacy vs. Technocratic Governance

Since NEPA's inception, two visions have competed for what environmental review is for. The first is a "Jeffersonian" or populist vision: NEPA as a democratic mechanism grounded in the premise that legitimate government requires an informed, participating citizenry. The second is NEPA as a technocratic process producing dense, specialized documents that Karkkainen describes as demanding "comprehensive bureaucratic rationality"—documents empirical research has found most ordinary citizens comprehend no better after reading than before (Karkkainen; Poisner, cited in Karkkainen). In practice, organized groups serve as intermediaries, repackaging EIS content for broader audiences, meaning what NEPA produces may be closer to interest group pluralism than to the direct democratic participation its proponents claim. This tension becomes acute in the permitting reform debate, where streamlining proposals risk resolving the conflict between efficiency and participation by simply eliminating participation—further concentrating the process in the hands of technical experts and developers while eroding the already limited access that affected communities have to decisions directly impacting their environments and health (Adelman et al.).

IV. NEPA's Actor-Network in Practice: Who Builds the Network, and How

A. Mapping the Actants

Federal agencies occupy the most structurally powerful position in the NEPA permitting network, functioning as focal actors that control the OPP: no major energy infrastructure project on federal land can proceed without their approval. Agencies such as the Bureau of Land Management, the Army Corps of Engineers, and the Forest Service each bring distinct institutional mandates and litigation-risk calculations to that gatekeeping role, and NEPA decision-making is shaped by "inherent subjectivity" in applying key concepts, giving field officers discretionary influence that varies by region and political context (Scott et al. 997).

The physical infrastructure—a pipeline route, a transmission corridor, a solar array—functions as a non-human actant whose material characteristics actively shape the network: where a project crosses federal land, which agencies are triggered, which statutory frameworks apply, and which communities enter the review process all depend in part on the project's geography (Adelman et al. 10040). The EIS document is similarly non-neutral: it structures which alternatives are considered feasible, which impacts count as significant, and which public comments receive substantive responses, shaping real-world outcomes well beyond its formal legal function (Mortimer et al. 2).

Project developers and their financial backers enter the network as enrollment-seekers, working to stabilize relationships with agencies, secure permits, and reduce the regulatory uncertainty that directly depresses investment. Investor behavior in the renewable energy sector is acutely sensitive to permitting timelines and policy instability; regulatory uncertainty propagates through contract negotiations and financing structures in ways that can halt investment entirely, irrespective of a project's underlying economics (Bolinger and Wiser 6). Indigenous nations occupy a structurally distinct and often resistant position in the network, legally entitled to government-to-government consultation under NEPA and the National Historic Preservation Act, yet maintaining that these consultation requirements are treated as procedural checkboxes rather than genuine enrollment. Environmental NGOs and federal courts function as counter-enrolling actants capable of destabilizing an otherwise assembled network: NGOs can mobilize NEPA's litigation provisions to challenge inadequate EIS analyses, while courts can vacate permits and force supplemental review—what Forest Service literature calls "process risk" (Mortimer et al. 3).

B. Translation in Action: The Dakota Access Pipeline Case

The Dakota Access Pipeline (DAPL) offers one of the clearest examples of how ANT's translation process unfolds and collapses within a NEPA permitting network. The focal actor, the United States Army Corps of Engineers (USACE), positioned statutory compliance as its OPP, the point through which all actants had to pass to legitimate the pipeline's construction. This framing successfully enrolled Dakota Access LLC, which prepared the draft EA on the USACE's behalf; courts, which ultimately deferred to the agency's legal determinations; and the Trump administration, which in February 2017 directed the USACE to proceed after the Obama administration had paused construction pending a full EIS (Johnson 337-38). The Standing Rock Sioux Tribe, however, were never meaningfully enrolled. Despite submitting 39 comments—more than any other individual commenter—and thoroughly documenting the pipeline's threats to their water supply, cultural sites, and treaty rights, the Tribe received dismissals rather than substantive engagement (Johnson 343). Thousands of people representing more than 300 Indigenous nations gathered at the Sacred Stone camp in what Johnson describes as the largest gathering of Indigenous people in modern history, demonstrating that the USACE had produced a network too narrow to hold (Johnson 338).

From an ANT perspective, the USACE's scoping process was not a neutral assessment of relevant impacts but an act of network boundary-drawing that determined which actants would be counted and which would be excluded. The USACE confined its environmental justice analysis to census tracts within the immediate construction zone, and when the Tribe argued the EA's geographic scope was too narrow to capture downstream risks to the reservation's water supply, the USACE simply reiterated the boundary—declaring that "the scope of this EA is limited to the proposed Project crossings of Corps-owned lands and flowage easements" and omitting further analysis because the nearest Standing Rock residence was 1.6 miles from the

construction site (U.S. Army Corps of Engineers, 2016a, cited in Johnson 344). The Tribe specifically argued that the EA "selectively uses and then changes the geographic areas by which it evaluates the impacts of the proposed pipeline . . . so that it effectively avoids the potential environmental impacts on the Tribe"—a charge the USACE never substantively addressed (Johnson 344). Tellingly, the original pipeline route had been redesigned away from Bismarck specifically because the USACE determined proximity to that city's water supply posed unacceptable risk, meaning the agency had already acknowledged the very kind of downstream harm it later refused to analyze for the Standing Rock Sioux (Johnson 346). In ANT terms, the EIS document's geographic scope functioned as a non-human actant that produced the Tribe's exclusion not by ignoring them, but by drawing the analytical boundary in a way that rendered their interests functionally invisible.

The USACE's engagement with the Tribe was a failure of *interessement*: rather than translating the Tribe's concerns about water rights, cultural sites, and treaty obligations into terms the permitting network could act on, the USACE treated those concerns as noise to be managed within the confines of legal compliance. Johnson argues this reflects a broader pattern in which "regulatory frameworks may exacerbate environmental justice concerns by incentivizing decision makers to prioritize justification for their decisions and avoiding legal battles over meaningfully engaging with communities" (Johnson 336). The Tribe's worldview—grounded in relational obligations to water, land, and future generations—does not map onto a census-tract-based environmental justice methodology, and translating it into network-compatible terms would have required restructuring the analysis itself, something the USACE had neither the mandate nor the incentive to do. This is not a failure attributable to any individual bad actor; it is a structural feature of a network whose OPP was defined as regulatory compliance rather than community consent.

C. The Mountain Valley Pipeline: Congressional Bypass as Enrollment

The Mountain Valley Pipeline (MVP) provision embedded in the FRA represented Congress acting as a macro-actant that forcibly enrolled every other node in the permitting network simultaneously. After more than five years of appellate reversals in which the Fourth Circuit had repeatedly vacated permits issued by the USACE, the Forest Service, the Fish and Wildlife Service, and state agencies, the FRA declared MVP to be in the national interest and directed all necessary agencies to issue required permits within three weeks (Smead 30). It also restricted judicial review to constitutional challenges heard exclusively in the D.C. Circuit, stripping the Fourth Circuit of the jurisdiction it had exercised to halt the project (Howe). As Richard Smead describes it, the provision functioned as a legislative *deus ex machina*: rather than resolving the network's instability through the translation process, Congress rewrote the rules of enrollment from the outside (Smead 30). Even so, the Fourth Circuit subsequently issued a stay of construction on an existing Forest Service permit after the FRA's passage, indicating that forced enrollment does not guarantee network stabilization (Smead 28).

The MVP provision was not the product of deliberative environmental policymaking; it was inserted late into debt-ceiling negotiations between President Biden and House Speaker Kevin McCarthy, rendering the pipeline's environmental record and affected communities' concerns subordinate to the political arithmetic of avoiding a national default (Smead 29-30). From an ANT perspective, this is a case where the OPP itself was relocated: the debt ceiling became the new OPP, and MVP's completion was the price of passage. What this reveals about NEPA is that the network it produces is not self-stabilizing; it depends on external institutional commitments to procedural integrity that a sufficiently powerful actant can dissolve, leaving affected communities suddenly without a forum.

V. The Justice Gap: Procedure, Substance, and Recognition in NEPA Review

A. What NEPA Does Well: Procedural Contributions

NEPA's most consistently recognized contribution is its transparency and disclosure architecture. In Ulibarri et al.'s multi-sector interview study, practitioners across all organization types identified disclosure as NEPA's defining strength: federal agency staff described it as making government action no longer "a secret"; NGOs identified it as the point at which the public gains access to government decision-making it would not otherwise have; and one state agency interviewee described NEPA as "probably one of the most democratic environmental statutes, if not the most democratic environmental statutes, that I'm aware of" (Ulibarri et al. 4). Karkkainen notes that NEPA's transparency, reinforced by judicial requirements that agencies respond substantively to public comments, can force agencies to surface information they would never have sought on their own—a meaningful epistemic contribution to governance even when the statute does not compel any particular outcome (Karkkainen 914). These procedural virtues have real value: they establish a public record, create accountability, and identify precisely where NEPA's architecture ends and where additional legal tools would need to begin.

B. The Substantive Limits of a Procedural Statute

NEPA's mandate ends at analysis. As a federal interviewee stated in Ulibarri et al.'s study, "at the end of your process and the record of decision . . . it's procedural and there's no real substantive mandate to mitigate," and this "toothless" quality was identified as the statute's most consequential limitation by practitioners across all organization types (Ulibarri et al. 5). Even when an EIS identifies significant disproportionate impacts on low-income or minority communities, the agency is under no legal obligation to select an alternative that avoids those impacts. Ulibarri et al. conclude that "NEPA's lack of substantive teeth fundamentally limits its ability to shape distributional outcomes," and that without an added mitigation requirement, the statute is structurally ill-equipped to reverse the patterns of inequitable siting the early environmental justice movement identified as central to the problem (Ulibarri et al. 9). California's Environmental Quality Act (CEQA), enacted the same year with a similar function, mandates mitigation rather than mere identification—a meaningful structural distinction. In

practice, however, CEQA still exhibits non-negligible environmental justice gaps: communities of color and already-polluted areas are statistically less likely to receive Mitigated Negative Declarations, suggesting that even a statute with substantive teeth produces contradictory distributional outcomes (Wang et al.).

C. NEPA's Notable Absence of Recognition Justice

NEPA's formal comment structure systematically disadvantages Indigenous and low-income communities: hearings during working hours, 60-day comment windows requiring expert witness preparation, and technically-dense documents create participation gaps that are structural rather than incidental. Ulibarri et al. find that recognition justice was nearly absent from practitioners' articulations of NEPA's goals—no interviewees described it as a central objective—and the law's procedural design targets a generic "public" rather than the specific communities most likely to bear disproportionate burdens (Ulibarri et al. 4-5). The DAPL case illustrates this directly: the Standing Rock Sioux Tribe's understanding of the Missouri River as a source of spiritual and cultural life was not translatable into the USACE's census-tract-based methodology, and when the Tribe attempted to articulate those concerns, they received jurisdictional deflection rather than substantive engagement (Johnson 343-44). Johnson argues this reflects a broader pattern in which regulatory frameworks define Indigenous groups as members of the general public for participation purposes, failing to account for the distinct treaty-based relationship between tribal nations and the federal government (Johnson 346). The FRA's streamlining provisions compound this exclusion by shortening comment timelines, narrowing the scope of required analysis to "reasonably foreseeable" impacts, and codifying limits on alternatives considered—reducing the surface area of the permitting process available for recognition-based claims precisely at the stages where they are most likely to arise (Clark, Kanatas & Smith 13).

VI. Technocracy, Reform, and the Green Energy Dilemma

A. The Permitting Reform Debate as a False Binary

The dominant narrative that NEPA is the primary obstacle to clean energy deployment rests on what Adelman et al. characterize as four interrelated myths, the most foundational of which is the misidentification of environmental review as the central bottleneck when the available empirical evidence points elsewhere (Adelman et al. 10039). In surveys of renewable energy developers, over 70% of solar developers and 65% of wind developers identified grid interconnection as the leading cause of delays exceeding six months; only 12-16% cited environmental restrictions, directly contradicting the legislative premise of the FRA's streamlining provisions (Adelman et al. 10050). Grid interconnection queues have grown to an average wait time of roughly 2.5 years nationally—driven not by environmental review but by unprecedented project volumes, rising grid upgrade costs, and sequential technical study requirements the current administrative structure was not designed to handle at this scale

(Adelman et al. 10050). Local zoning ordinances, identified in a 2022 study as more than 1,800 restrictive ordinances for wind and 800 for solar, could reduce available resources by as much as 87% and 38% respectively—barriers no amount of NEPA streamlining can address (Adelman et al. 10050). Texas, where virtually no projects require NEPA review and interconnection is comparatively fast, still sees solar projects taking three to five years to develop, demonstrating that NEPA's procedural requirements are neither necessary nor sufficient to explain the timelines reform advocates cite as evidence of the law's burden (Adelman et al. 10051). Framing NEPA as the obstacle converts fundamentally political conflicts over the distribution of infrastructure costs and community consent into a narrative about procedural inefficiency, directing reform energy toward restricting public participation rather than the grid investment, agency resourcing, and inter-jurisdictional coordination that the evidence actually identifies as binding constraints.

B. What Reform Gets Wrong

The current push for permitting reform embodies a technocratic logic that treats speed and efficiency as ends in themselves, framing community opposition and procedural requirements as frictions to be eliminated rather than as legitimate expressions of democratic conflict over who bears the costs of infrastructure development. Adelman et al. characterize deregulatory approaches such as "arbitrary limits on NEPA processes, exemptions from environmental review, and eliminating agency staff" as "fundamentally misconceived and misdirected," warning that "hasty permitting reforms, especially those aimed at NEPA, risk eliminating public involvement" with the result that "environmental justice and tribal communities will be disadvantaged" (Adelman et al. 10055). When Congress ordered the Mountain Valley Pipeline into existence through the FRA, it did not solve the underlying conflict over the pipeline's impacts on affected communities—it simply removed those communities' remaining procedural leverage by stripping the Fourth Circuit of jurisdiction and compressing agency action into a three-week window (Smead 30). The FRA's narrowing of required analysis to "reasonably foreseeable" impacts and its restriction of alternatives to those "technically and economically feasible" do not reduce the actual environmental and distributional conflicts NEPA review surfaces; they limit the portion of the process in which affected communities can raise those conflicts, effectively resolving political disputes through administrative redefinition (Clark, Kanatas & Smith 13-14). The historical record provides a cautionary baseline: Adelman et al. note that the Interstate Highway System—often cited as a model of rapid infrastructure deployment—destroyed communities of color and low-income neighborhoods precisely because it operated without the public processes that permitting reformers now seek to curtail (Adelman et al. 10048).

VII. Conclusion

NEPA was designed to make the environmental consequences of federal action legible before decisions are made and to create space for democratic participation in those decisions. As this paper has argued, it succeeds at the first task more than the second, and at neither to the degree

its most ardent defenders claim. Analyzed through ANT and the three-dimensional environmental justice framework, NEPA's permitting network emerges not as a neutral administrative process but as a sociotechnical system that enrolls some actants and excludes others in ways that are structural rather than incidental.

The DAPL case makes this concrete. The USACE's OPP was defined as regulatory compliance, not community consent, and the Tribe's relational obligations to the Missouri River could not be translated into the EA's census-tract-based methodology. The EIS document's geographic scope—a non-human actant—drew the analytical boundary in a way that rendered the Tribe's interests functionally invisible. This is a recognition justice failure at the network level: the permitting process had no mechanism for accommodating the Tribe's worldview, and no incentive to develop one. The MVP provision reveals a complementary vulnerability: when a sufficiently powerful actant chooses to redefine the OPP, the procedural infrastructure through which communities had organized their resistance is dismantled by legislative fiat, leaving those communities without a forum regardless of the merits of their concerns.

The permitting reform debate, as currently framed, offers no solution to either problem. The empirical evidence is clear that NEPA review is not the primary cause of renewable energy delays—grid interconnection backlogs, local zoning restrictions, and chronically underfunded agencies are. Streamlining environmental review without addressing these structural constraints produces neither faster energy deployment nor more just outcomes. It produces a permitting process more insulated from democratic challenge, which may explain a significant portion of its political appeal.

Meaningful reform would require redefining the network's OPP: moving from a standard of regulatory compliance toward one of substantive community engagement. This would mean earlier public participation before project designs are finalized, mitigation requirements rather than identification-only mandates, and genuine recognition of tribal nations' distinct treaty-based status rather than their treatment as members of the general public. The CEQA comparison is instructive as a partial model, though Wang et al.'s empirical findings caution that even a statute with substantive teeth produces contradictory distributional outcomes in practice, suggesting that structural reform must reach further than any single procedural fix. NEPA's translation failures are, in the end, failures of network design. The statute built a process that is good at producing documents and poor at producing justice. Reforming it in ways that accelerate the former while eroding the conditions for the latter is not a solution to the energy transition's challenges—it is a way of ensuring that the costs of that transition continue to fall on the communities least able to bear them.

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